

Comparative Evaluation of Advanced Gradient Boosting Models for Predicting Salt Adsorption Capacity of Faradic Materials

Ishan Jain
240968430

Yash Agarwal
240968502

Pratham R Shenoy
220968398

ishan5.mitmpl2024@learner.manipal.edu yash12.mitmpl2024@learner.manipal.edu pratham8.mitmpl2022@learner.manipal.edu
MIT, Manipal MIT, Manipal MIT, Manipal

Abstract—Accurate prediction of salt adsorption capacity (SAC) can reduce the experimental burden of screening faradic electrode materials for capacitive deionization (CDI). A recent data-driven study predicted SAC from specific capacitance (Cs), specific surface area (SSA), electrolyte concentration (EC), and applied voltage (AV), reporting gradient boosting decision tree (GBDT) errors of 6.34 mg g^{-1} for mean absolute error (MAE) and 8.58 mg g^{-1} for root mean square error (RMSE) [25]. We re-examined the same 510-sample, four-feature task in two complementary investigations. Study 1 followed a fixed 80:20 train-test framework and evaluated XGBoost, LightGBM, CatBoost, and NGBoost before and after five-fold grid-search tuning. Tuned CatBoost achieved the lowest held-out errors (MAE 5.110 mg g^{-1} ; RMSE 7.502 mg g^{-1}), representing indicative reductions of 19.4% and 12.6%, respectively, relative to the values reported for GBDT. Study 2 extended the analysis through nested cross-validation: three-fold inner tuning was conducted within each of five outer folds, producing one out-of-fold prediction for every sample across eight algorithms. XGBoost ranked first in this more comprehensive evaluation (MAE 5.2252 mg g^{-1} ; RMSE 7.9470 mg g^{-1} ; R^2 0.8649), reducing pooled OOF RMSE by 4.79% relative to the GBDT baseline evaluated under the same protocol. SHAP analysis identified EC and Cs as the dominant CatBoost features in Study 1 and Cs and EC as the dominant XGBoost features in Study 2. The close reversal of their ordering shows that strong predictors can retain importance while their exact rank varies with model and validation design. These results support modern boosting for SAC prediction while emphasizing validation-aware interpretation and the need for grouped and prospective experimental confirmation.

Index Terms—capacitive deionization; faradic materials; salt adsorption capacity; gradient boosting; nested cross-validation; out-of-fold prediction; SHAP

I. INTRODUCTION

Freshwater scarcity and deteriorating source-water quality have intensified the need for efficient desalination. Reverse osmosis and electrodialysis are mature technologies, but their energy and operational requirements motivate complementary approaches [1,2]. Capacitive deionization removes ions through electrically driven adsorption and electrochemical storage, offering a potentially energy-efficient route for brackish-water treatment [3]. Its performance depends strongly on electrode chemistry, accessible ion-storage sites, cell architecture, and operating conditions.

Conventional porous carbon electrodes offer high surface area and electrical conductivity but often have limited charge-storage capacity. Faradic materials exploit redox reactions, intercalation, or pseudocapacitive processes to increase ion uptake [4,5]. Prussian blue analogues [6,7], defect-rich molybdenum disulfide [8], rocking-chair architectures [9], and metal-organic-framework-derived electrodes [10] illustrate the expanding materials space. Reviews of metal sulfides further show that composition, defects, and structure can strongly affect CDI performance [11]. Exploring these combinations experimentally remains expensive and time-consuming.

Machine learning (ML) provides a complementary route for relating material descriptors and operating variables to performance. Data-driven approaches have been used for battery electrodes, porous carbons, electrolytes, and energy materials [12–18]. Within CDI, ML has been applied to feature-contribution analysis, electrode-capacity prediction, membrane-process modelling, metadata analysis, and porous-carbon performance prediction [19–23]. The broader role of ML in CDI has consequently become an active research direction [24].

Wang et al. [25] assembled 510 observations of faradic materials and predicted SAC using Cs, SSA, EC, and AV. Among GBDT, random forest (RF), support vector machine (SVM), and artificial neural network (ANN) models, GBDT produced the lowest reported errors (MAE 6.34 mg g^{-1} and RMSE 8.58 mg g^{-1}). SHapley Additive exPlanations (SHAP) ranked Cs as the most influential feature, and the reported trends were examined experimentally using Prussian blue analogues and MoS_2 -based electrodes.

The present computational study asks whether advanced boosting frameworks and systematic tuning improve this prediction task and whether conclusions remain stable under a more comprehensive validation design. Study 1 follows the principal fixed-split methodology of the antecedent study and compares XGBoost, LightGBM, CatBoost, and NGBoost. Study 2 is an additional robustness analysis that compares those models with GBDT, RF, SVM, and ANN using nested cross-validation and pooled out-of-fold (OOF) predictions. Interpretability follows established applications of interpretable

TABLE I
DATASET AND EVALUATION SUMMARY.

Property	Value
Observations	510
Predictors	Cs, SSA, EC, AV
Target	SAC (mg g ⁻¹)
Missing values	0
Exact duplicate rows	0
Study 1 split	408 training; 102 testing

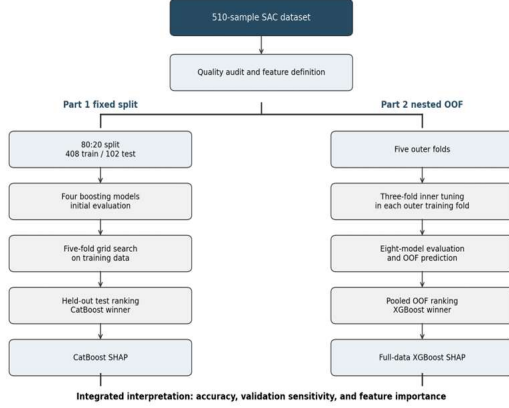


Fig. 1. Overall computational framework. Study 1 uses a fixed 80:20 split and identifies CatBoost as the held-out test winner. Study 2 uses nested five-fold OOF evaluation and identifies XGBoost as the aggregate winner.

machine learning and feature-attribution methods in battery-health, biochar, electrocatalytic-material, porous-carbon, and CDI studies [26–30]. The contribution is therefore both predictive and methodological: it identifies the winner under each protocol and demonstrates how validation design affects model ranking and feature interpretation.

II. MATERIALS AND METHODS

A. Dataset and variables

The supplied dataset contains 510 observations and five numeric columns. Four predictors were used: Cs (dataset column C, F g⁻¹), SSA (m² g⁻¹), EC (mg L⁻¹), and AV (V). SAC (mg g⁻¹) was the regression target. These variables match the antecedent task [25]. No missing values or exact duplicate rows were detected. Exact-row screening does not establish independence between measurements from the same material or publication; this distinction is addressed under limitations.

B. Overall computational framework

The analysis separates the reference-aligned fixed-split experiment from the additional nested OOF experiment (Fig. 1). This separation preserves the scientific meaning of both outcomes: Study 1 identifies the best model on a designated test partition, whereas Study 2 estimates performance across all observations through predictions generated without training on the predicted row.

C. Study 1 fixed-split evaluation

The data were randomly divided into training (80%, $n = 408$) and testing (20%, $n = 102$) subsets using `random_state = 42`. XGBoost, LightGBM, CatBoost, and NGBoost were first fitted using the initial configurations recorded in the notebook. Initial-model performance within the training subset was evaluated using shuffled five-fold cross-validation with `random_state = 42`. Model-specific hyperparameters were subsequently optimized on the same training subset using five-fold GridSearchCV. In the supplied implementation, GridSearchCV(`cv=5`) used the default unshuffled five-fold splitter. Neither cross-validation procedure accessed the 102-observation test subset. Each selected estimator was refitted using all 408 training observations and evaluated on the 102 held-out observations. Because all Study 1 models were tree-based ensembles, feature scaling was not applied.

D. Study 2 nested out-of-fold evaluation

Study 2 compared GBDT, RF, SVM, ANN, XGBoost, LightGBM, CatBoost, and NGBoost. A shuffled five-fold outer cross-validation partition was created with `random_state = 42`. For every model and outer fold, a separate three-fold shuffled inner GridSearchCV optimized hyperparameters using only the outer training subset. The refitted inner-CV winner then predicted the untouched outer validation fold. Repetition across five folds produced one OOF prediction for each of the 510 observations. StandardScaler was contained inside the SVM and ANN pipelines, ensuring that scaling parameters were learned without access to held-out rows.

$$\hat{y}_i^{\text{OOF}} = f_{-k(i)}(\mathbf{x}_i), \quad i \in I_{k(i)} \quad (1)$$

Here, $I_{k(i)}$ denotes the outer fold containing observation i and $f_{-k(i)}$ denotes the tuned estimator trained without that fold. Pooled metrics were calculated after restoring predictions to their original row positions.

E. Models and hyperparameter optimization

XGBoost uses regularized second-order gradient boosting; LightGBM grows trees leaf-wise; CatBoost uses ordered boosting and symmetric trees; and NGBoost optimizes a probabilistic predictive distribution through natural gradients. Only NGBoost point predictions were used for the reported regression metrics. Study 2 additionally included the four model families studied by Wang et al. [25]. The grid-search spaces and selected settings are reported in the main results and appendices. The Study 2 grids were intentionally compact to make nested tuning computationally tractable.

F. Evaluation metrics

Study 1 retained the antecedent study’s MAE, MSE, and RMSE metrics. Study 2 additionally reported the coefficient of determination, R^2 . Lower MAE, MSE, and RMSE indicate

TABLE II
STUDY 1 TEST PERFORMANCE FOR THE INITIAL MODEL CONFIGURATIONS.

Model	MAE	MSE	RMSE
XGBoost	5.816	65.269	8.079
NGBoost	6.845	80.084	8.949
CatBoost	6.924	83.767	9.152
LightGBM	7.097	92.397	9.612

TABLE III
STUDY 1 FIVE-FOLD CROSS-VALIDATION RESULTS ON THE TRAINING SUBSET.

Model	Mean CV RMSE	SD
XGBoost	8.794	1.720
NGBoost	9.901	1.430
CatBoost	10.258	1.349
LightGBM	11.999	1.566

smaller prediction error, while higher R^2 indicates greater explained target variance.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (5)$$

G. SHAP analysis

TreeExplainer was applied to the tuned CatBoost model on the Study 1 test subset. Following the nested OOF evaluation in Study 2, a separate XGBoost model was fitted to all 510 observations for post-hoc interpretation. This model used the modal optimal configuration selected by the compact inner grid searches across the five outer folds: 300 estimators, learning rate 0.1, maximum depth 4, row subsampling 0.8, and column subsampling 0.8. All five folds selected the same settings except maximum depth, for which four folds selected 4 and one fold selected 3. The full-data model was used only for SHAP interpretation; all reported Study 2 performance metrics were calculated exclusively from outer-fold OOF predictions. Mean absolute SHAP values summarize global model-attributed influence, while summary and dependence plots show direction and heterogeneity. SHAP explains the fitted prediction function; it does not establish physical causation [26,27,30].

III. RESULTS AND DISCUSSION

A. Study 1 initial performance

Among the initial model configurations, XGBoost achieved the lowest held-out error (MAE 5.816 mg g⁻¹; RMSE 8.079 mg g⁻¹), followed by NGBoost, CatBoost, and LightGBM (Table II). The initial five-fold training CV also ranked XGBoost first, with a mean RMSE of 8.794 mg g⁻¹ (Table III).

TABLE IV
STUDY 1 HYPERPARAMETERS SELECTED BY GRID SEARCH AND CORRESPONDING TRAINING CV RMSE.

Model	Selected hyperparameters	CV RMSE
XGBoost	500 trees; lr 0.1; depth 5; subsample 0.8; columns 1.0	7.840
NGBoost	500 estimators; lr 0.1; minibatch 0.8	8.310
CatBoost	500 iterations; lr 0.1; depth 6; L2 1	8.683
LightGBM	300 trees; lr 0.1; depth 5; 15 leaves; min child 5	9.282

TABLE V
STUDY 1 TUNED-MODEL PERFORMANCE ON THE HELD-OUT TEST SUBSET.

Model	MAE	MSE	RMSE
Tuned CatBoost	5.110	56.282	7.502
Tuned XGBoost	5.524	58.596	7.655
Tuned LightGBM	5.384	64.679	8.042
Tuned NGBoost	5.675	65.534	8.095

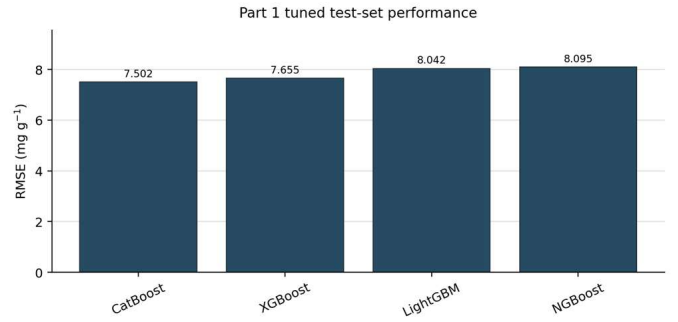


Fig. 2. Study 1 test RMSE after hyperparameter tuning. CatBoost achieved the lowest error on the fixed test partition.

B. Study 1 tuning and held-out test performance

Training-set grid search selected XGBoost as the lowest CV-error candidate, but final held-out evaluation produced a narrow ranking reversal. Tuned CatBoost achieved the lowest test MAE and RMSE, and is therefore the Study 1 winner under the fixed-split comparison (Table V and Fig. 2). XGBoost remained close, with RMSE only 0.153 mg g⁻¹ higher. This does not invalidate either result: the CV score summarizes resampled training performance, whereas the test score describes generalization to the designated 102 observations.

C. Study 1 comparison with the antecedent study

Relative to the GBDT values reported by Wang et al. [25], the tuned CatBoost result is numerically lower by 19.4% for MAE and 12.6% for RMSE (Table VI). The comparison is informative because dataset size, features, target, and broad evaluation structure are shared. It remains indicative rather than a controlled ablation because the exact reference test indices, software versions, and tuning implementation were not available for verification.

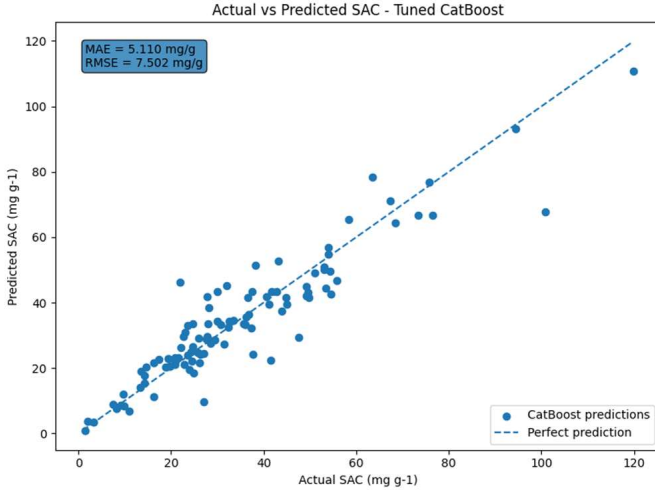


Fig. 3. Actual versus predicted SAC for tuned CatBoost on the Study 1 test subset. The diagonal denotes perfect prediction.

TABLE VI
INDICATIVE COMPARISON OF STUDY 1 CATBOOST WITH THE GBDT
RESULT REPORTED BY WANG ET AL. [25].

Study and model	MAE	RMSE	Interpretation
Wang et al. [25] GBDT	6.34	8.58	Reported reference result
This work: tuned CatBoost	5.11	7.50	Study 1 fixed-split result
Indicative reduction	19.4%	12.6%	Not a same-index ablation

TABLE VII
STUDY 1 TUNED CATBOOST GLOBAL SHAP IMPORTANCE ON THE TEST
SUBSET.

Feature	Mean SHAP
EC	7.451
Cs	7.051
SSA	5.027
AV	3.922

D. Study 1 CatBoost interpretation

For tuned CatBoost, EC produced the largest mean absolute SHAP value (7.451), closely followed by Cs (7.051); SSA and AV were less influential (Table VII and Fig. 4). The Cs dependence plot was broadly positive (Fig. 5), consistent with its physical role as an indicator of charge and ion-storage capability [19,20,25]. The reversal from the Cs-first ranking reported by Wang et al. [25] is small in magnitude and suggests sensitivity to model structure and fitted interactions rather than a contradiction in underlying electrochemical plausibility.

E. Study 2 nested OOF performance

Nested OOF evaluation produced a different and more comprehensive ranking (Table VIII). XGBoost obtained the

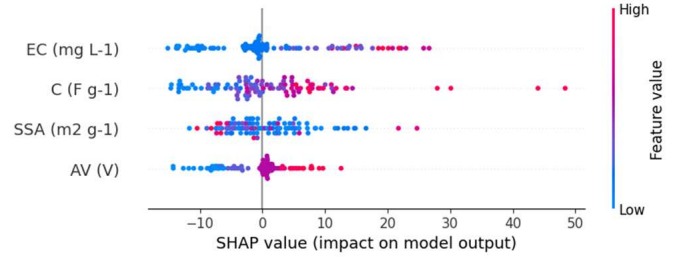


Fig. 4. SHAP summary plot for tuned CatBoost on the Study 1 test subset.

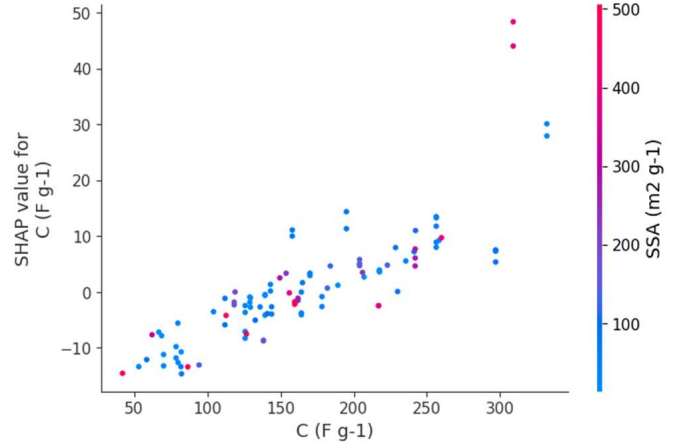


Fig. 5. CatBoost SHAP dependence plot for specific capacitance in Study 1.

TABLE VIII
STUDY 2 POOLED NESTED OOF PERFORMANCE ACROSS ALL 510
OBSERVATIONS.

Rank	Model	MAE	MSE	RMSE	R^2
1	XGBoost	5.2252	63.1548	7.9470	0.8649
2	GBDT	5.4952	69.6741	8.3471	0.8509
3	NGBoost	6.0068	72.5690	8.5187	0.8448
4	CatBoost	6.4314	81.8778	9.0486	0.8248
5	LightGBM	7.0294	97.1449	9.8562	0.7922
6	Random Forest	6.9188	100.3967	10.0198	0.7852
7	ANN	9.2566	159.0663	12.6121	0.6597
8	SVM	11.3066	307.3888	17.5325	0.3424

lowest observed pooled errors (MAE 5.2252 mg g⁻¹; RMSE 7.9470 mg g⁻¹) and the highest observed R^2 (0.8649) under the specified nested OOF protocol. GBDT ranked second, followed by NGBoost and CatBoost. XGBoost reduced RMSE by 4.79% relative to the GBDT trained and evaluated under the same nested OOF protocol. This comparison is controlled within Study 2 and should not be confused with the separately reported GBDT result in [25].

F. Study 2 stability and model comparison

XGBoost produced a lower RMSE than GBDT in three of the five outer folds, whereas GBDT produced a lower RMSE in the remaining two folds. The fold-level results therefore indicate a modest and fold-dependent difference rather than a universal advantage. The pooled result nevertheless

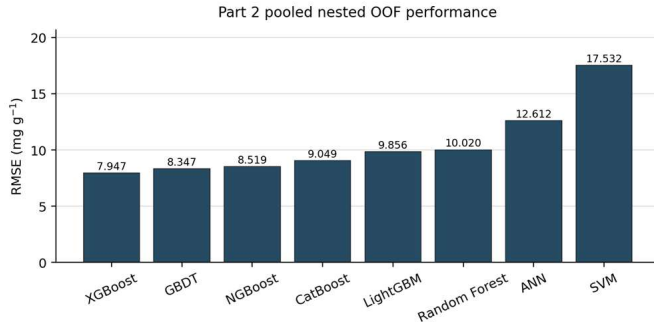


Fig. 6. Study 2 pooled nested OOF RMSE. XGBoost achieved the lowest aggregate error.

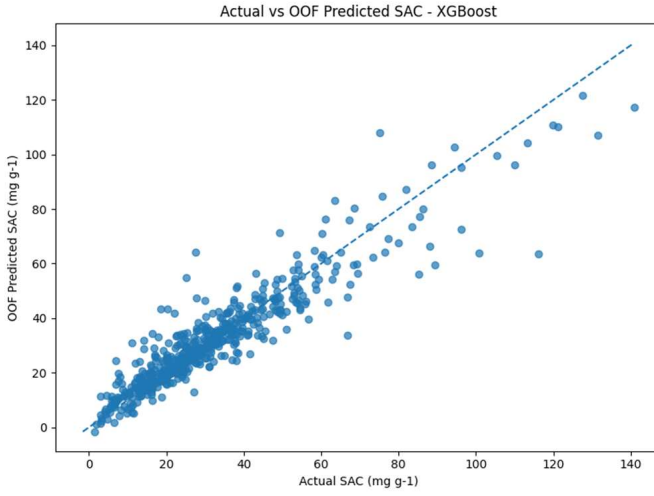


Fig. 7. Actual versus OOF-predicted SAC for XGBoost in Study 2.

TABLE IX
FOLD-LEVEL RMSE COMPARISON BETWEEN THE TWO LEADING STUDY 2 MODELS.

Fold	XGBoost RMSE	GBDT RMSE	Lower error
1	8.0421	8.5405	XGBoost
2	9.4254	11.4416	XGBoost
3	9.2459	8.2055	GBDT
4	5.6398	6.3757	XGBoost
5	6.7056	6.0449	GBDT

integrates all 510 held-out predictions and is less dependent on one designated test subset. Row-level win counts provided supporting descriptive evidence: XGBoost had the smallest absolute error for 110 observations (21.57%), followed by GBDT for 89 (17.45%). These counts were not used as the primary selection rule because they do not reflect the magnitude of losses on non-winning observations.

G. Study 2 XGBoost interpretation

The final full-data XGBoost interpretation model ranked Cs first (mean |SHAP| 8.183) and EC second (8.046), followed by SSA and AV (Table X and Fig. 8). The near equality of Cs and EC reinforces the cross-study conclusion that both are dominant predictors. Dependence plots for Cs and EC (Figs. 9

TABLE X
STUDY 2 GLOBAL SHAP IMPORTANCE FOR THE FINAL FULL-DATA XGBOOST MODEL.

Feature	Mean SHAP
Cs	8.183
EC	8.046
SSA	5.177
AV	4.285

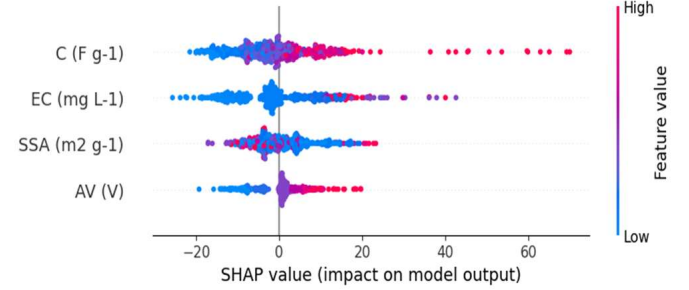


Fig. 8. SHAP summary plot for the final XGBoost interpretation model fitted to all 510 observations.

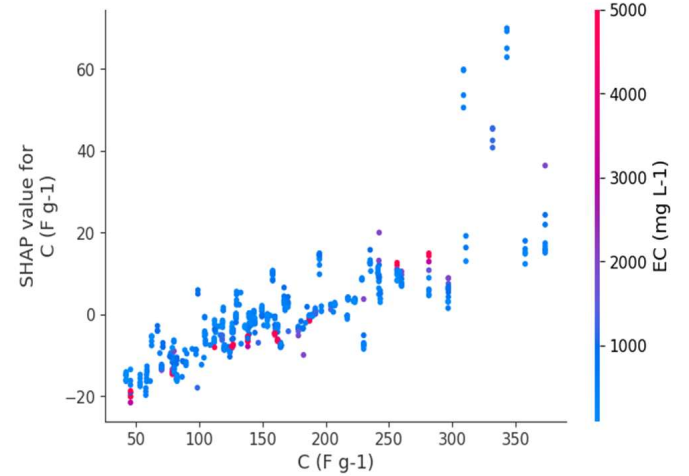


Fig. 9. XGBoost SHAP dependence plot for specific capacitance after Study 2.

and 10) should be interpreted as fitted associations, not isolated causal effects, particularly because correlated predictors can redistribute SHAP attribution.

H. Integrated interpretation

The different winners answer different questions. CatBoost obtained the lowest observed error on the fixed Study 1 test subset and provides the principal comparison with the antecedent paper's methodology. XGBoost obtained the lowest observed aggregate error when all observations were evaluated out of fold under nested tuning. XGBoost also led the Study 1 training CV, making its Study 2 success coherent with the broader resampling evidence. The conclusion is not that one result cancels the other, but that rankings among competitive

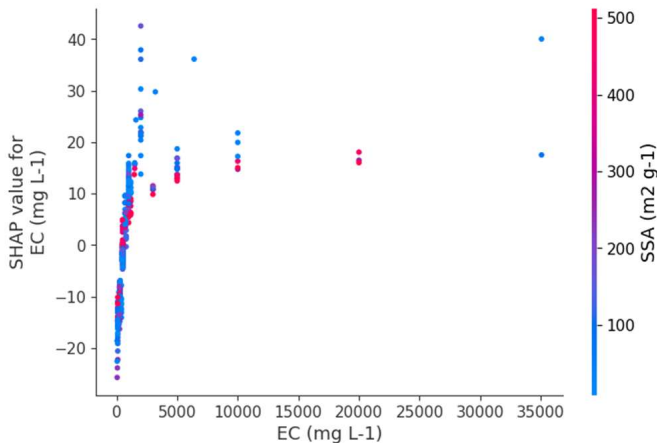


Fig. 10. XGBoost SHAP dependence plot for electrolyte concentration after Study 2.

boosting models can change with validation design and sample composition.

The interpretability results show a parallel pattern. Study 1 places EC marginally above Cs, whereas Study 2 and Wang et al. [25] place Cs first. Across all analyses, however, Cs and EC occupy the leading positions. This stable pair is more scientifically defensible than a claim of universal dominance by either variable. Cs describes the electrode’s charge-storage capability, while EC controls ion availability; both plausibly govern SAC, and nonlinear interactions with SSA and AV may alter their marginal model attribution [19,20,25,30].

IV. LIMITATIONS AND FUTURE WORK

Study 1 relies on one fixed test partition, and the exact test indices used by the antecedent paper were not available; its percentage improvements are therefore indicative. Study 2 reduces split dependence through nested OOF prediction but uses one five-fold outer partition. Repeated nested CV and bootstrap confidence intervals would quantify ranking uncertainty more fully. Nested tuning protected each outer-fold evaluation from hyperparameter-selection leakage within a model family. However, the same pooled OOF results were subsequently used to compare eight model families and identify the model with the lowest observed error. Consequently, the reported XGBoost OOF error should be interpreted as a comparative performance estimate from the present evaluation rather than as a completely unbiased estimate of the future performance of a finally selected model family. Repeated nested cross-validation or evaluation on an independent external dataset would provide stronger confirmation of the final model ranking. Several selected hyperparameters occur at search-grid boundaries, and the compact Study 2 grids are not equally broad across algorithms. CatBoost, for example, was limited to fewer iterations and shallower trees in Study 2 than in Study 1, so the nested ranking applies to the specified search spaces. ANN generated convergence warnings in some folds and could benefit from revised optimization settings.

The dataset audit found no exact duplicates, but row-wise splitting may still distribute related observations from the same material or publication across training and validation folds. If material and source identifiers become available, grouped validation should be prioritized to measure transfer to genuinely unseen material families. Finally, SHAP values describe model behavior rather than causality, and the present work did not synthesize materials or conduct CDI experiments. Prospective tests on newly selected faradic electrodes remain necessary before model predictions can guide deployment. Because no paired confidence interval or formal hypothesis test was performed, differences between the models are reported descriptively and should not be interpreted as evidence of statistically significant superiority.

V. CONCLUSIONS

Two complementary evaluations were conducted for SAC prediction from Cs, SSA, EC, and AV. In Study 1, tuned CatBoost achieved the best performance on the fixed 20% test subset, with MAE 5.110 mg g^{-1} and RMSE 7.502 mg g^{-1} . These values were numerically 19.4% and 12.6% lower, respectively, than the GBDT errors reported by Wang et al. [25], subject to the limitation that exact reference split indices were unavailable. In Study 2, XGBoost obtained the lowest observed pooled OOF error among the eight evaluated algorithms, with MAE 5.2252 mg g^{-1} , RMSE 7.9470 mg g^{-1} , and R^2 0.8649. It reduced RMSE by 4.79% relative to the GBDT baseline evaluated under the identical nested protocol. The different winners demonstrate the sensitivity of model ranking to validation design rather than an inconsistency. SHAP analyses consistently identified Cs and EC as the two dominant predictors, although their exact ordering changed. Modern boosting methods therefore offer improved SAC prediction on this dataset, while robust validation, grouped testing, and prospective experiments remain essential for materials-design claims.

DATA AND CODE AVAILABILITY

The dataset and computational notebooks supporting the reported analyses accompany this study. A public repository or permanent identifier has not yet been assigned.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX A

STUDY 1 COMPUTATIONAL MAP

The fixed-split workflow was: (1) load 510 observations; (2) verify schema, missingness, exact duplicates, and numeric types; (3) define $X = \{\text{Cs, SSA, EC, AV}\}$ and $y = \text{SAC}$; (4) create the 408/102 split using random_state 42; (5) fit four initial models; (6) calculate held-out MAE, MSE, and RMSE; (7) perform five-fold training CV; (8) define model-specific grids; (9) tune only on the training subset; (10) refit each

selected estimator to all 408 training observations; (11) predict the 102 untouched test observations; (12) calculate final metrics; (13) rank by RMSE; (14) designate CatBoost as the Study 1 test winner; and (15) calculate CatBoost SHAP values on the test subset.

Algorithm A1: $D \rightarrow \text{split}(D, 0.8, \text{seed } 42) \rightarrow \text{inner CV tuning} \rightarrow \text{refit} \rightarrow \text{test prediction} \rightarrow \text{metrics} \rightarrow \text{CatBoost SHAP}$

APPENDIX B

STUDY 2 NESTED OOF COMPUTATIONAL MAP

For each model m and each outer fold k , the outer validation indices I_k were withheld. Three-fold GridSearchCV operated only on $D \setminus I_k$. The best inner configuration was refitted to $D \setminus I_k$ and predicted I_k . Predictions were stored at their original row indices. After five folds, every row contained one OOF prediction for model m ; pooled metrics were then calculated and models ranked. This procedure prevents both preprocessing and hyperparameter selection from accessing an observation before its OOF prediction is produced.

For m in models: for k in $1 \dots 5$: $\theta_{mk}^* = \arg \min_{\theta} \text{CV}_3(D \setminus I_k)$; fit $f_{mk}(\theta_{mk}^*)$; predict I_k ; pool $\hat{y}^{\text{OOF}}$

APPENDIX C

MATHEMATICAL BASIS OF CATBOOST

CatBoost represents the prediction as an additive ensemble $F_t(x) = F_{t-1}(x) + \eta h_t(x)$, where h_t is a newly fitted symmetric tree and η is the learning rate. At iteration t , the tree reduces the gradient of the selected loss with respect to current predictions. Symmetric trees apply the same split condition across all nodes at a given depth, creating regular structure and limiting arbitrary partition growth. Ordered boosting estimates residual-like quantities using preceding observations in an internal permutation, reducing prediction shift. In this project all inputs were numerical, so CatBoost’s handling of categorical variables was not the central advantage. Its Study 1 performance instead reflects the combined effects of ordered boosting, symmetric-tree regularization, depth 6, 500 iterations, learning rate 0.1, and L2 leaf regularization of 1.

APPENDIX D

MATHEMATICAL BASIS OF XGBOOST

$$\mathcal{L}^{(t)} = \sum_i l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \quad (6)$$

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (7)$$

$$\mathcal{L}^{(t)} \approx \sum_i [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (8)$$

Here g_i and h_i are the first and second derivatives of the loss with respect to the previous prediction, T is the number of leaves, and w contains leaf weights. XGBoost therefore combines sequential correction with curvature information and explicit complexity penalties. For the final full-data interpretation model, the modal inner-CV-optimal configuration across the five outer folds was used: row subsampling 0.8, column subsampling 0.8, maximum depth 4, learning rate 0.1,

TABLE A-I
STUDY 1 HYPERPARAMETER SEARCH SPACES.

Model	Study 1 search space
XGBoost	trees 100/200/300/500; lr .01/.05/.1; depth 2–5; rows .8/1.0; columns .8/1.0
LightGBM	trees 100/200/300; lr .01/.05/.1; depth 2–5; leaves 7/15/31; child 5/10/20
CatBoost	iterations 100/200/300/500; lr .01/.05/.1; depth 3–6; L2 1/3/5/10
NGBoost	estimators 100/200/300/500; lr .01/.05/.1; minibatch 1.0/.8

and 300 estimators. Maximum depth 4 was selected in four of the five outer-fold grid searches, while all other listed settings were selected in every fold. This final refit was used exclusively for SHAP analysis and did not contribute to the nested OOF performance estimates. These mechanisms are well suited to nonlinear tabular relationships and may have contributed to XGBoost obtaining the lowest observed pooled OOF error.

APPENDIX E

METRIC AND IMPROVEMENT VERIFICATION

Study 1 CatBoost MAE reduction = $(6.34 - 5.110)/6.34 \times 100 = 19.4\%$. Study 1 CatBoost RMSE reduction = $(8.58 - 7.502)/8.58 \times 100 = 12.6\%$. Study 2 XGBoost-versus-GBDT OOF RMSE reduction = $(8.3471 - 7.9470)/8.3471 \times 100 = 4.79\%$. Study 1 follows the antecedent paper by emphasizing MAE, MSE, and RMSE; R^2 is additionally reported in Study 2.

APPENDIX F

SHAP BACKGROUND

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f_{S \cup \{j\}}(x) - f_S(x)] \quad (9)$$

The local value ϕ_j attributes a prediction difference to feature j by averaging its marginal contribution across feature subsets. The sign indicates whether a feature contribution raises or lowers the prediction relative to the explainer baseline. Mean $|\phi_j|$ across observations provides a global importance ranking but discards direction. Dependence plots retain direction and can reveal nonlinearity or interactions. When predictors are correlated, attribution can be redistributed among them; neither local nor global SHAP values establish causality.

APPENDIX G

HYPERPARAMETER AND REPRODUCIBILITY RECORD

Random seeds were fixed at 42. The preliminary Study 1 cross-validation used shuffled folds, whereas the subsequent GridSearchCV(cv=5) searches used the default unshuffled five-fold splitter. Study 2 used five shuffled outer folds and three shuffled inner folds. Scaling was contained in SVM and ANN pipelines. The ANN optimizer reached its 500-iteration maximum in some fits. Exact package versions were not recorded in the supplied notebook; these should be

TABLE A-II
STUDY 2 COMPACT NESTED-SEARCH SPACES.

Model	Study 2 compact search space
GBDT	trees 100/200/300; lr .05/.1; depth 2/3/4
RF	trees 200/400; depth None/9; features sqrt/1.0
SVM	C 1/3/10; gamma scale/.1; epsilon .1
ANN	layers 50, 100, or 50–25; lr .001/.005
XGBoost	trees 100/200/300; lr .05/.1; depth 2/3/4; rows .8; columns .8
LightGBM	trees 100/200; lr .05/.1; depth 3/5; leaves 15
CatBoost	iterations 100/200; lr .05/.1; depth 4/5; L2 3
NGBoost	estimators 100/200; lr .05/.1; minibatch .8

archived from the original execution environment in the final reproducibility package.

APPENDIX H AUDIT TRAIL

Verification map: raw CSV → numeric feature matrix and SAC vector → fixed split or nested fold indices → training-only tuning → fitted estimator → held-out predictions → fold and pooled metrics → model ranking → SHAP interpretation. Required checks are: 510 source observations; 510 populated OOF predictions for each Study 2 model; no missing OOF predictions; each row predicted once per model; no outer validation row used by its inner grid search; consistent units; rankings reproduced from notebook outputs; and percentage improvements recalculated from unrounded errors.

REFERENCES

- [1] S. Al-Amshawee, M.Y.B. Yunus, A.A.M. Azodein, D.G. Hassell, I.H. Dakhil, H. Abu Hasan, “Electrodialysis desalination for water and wastewater: a review,” *Chemical Engineering Journal*, vol. 380, 122231, 2020. DOI: <https://doi.org/10.1016/j.cej.2019.122231>
- [2] M. Qasim, M. Badrelzaman, N.N. Darwish, N.A. Darwish, N. Hilal, “Reverse osmosis desalination: a state-of-the-art review,” *Desalination*, vol. 459, pp. 59–104, 2019. DOI: <https://doi.org/10.1016/j.desal.2019.02.008>
- [3] K. Sun, M. Tebyetekerwa, C. Wang, X. Wang, X. Zhang, X.S. Zhao, “Electrocapacitive deionization: Mechanisms, electrodes, and cell designs,” *Advanced Functional Materials*, vol. 33, 2213578, 2023. DOI: <https://doi.org/10.1002/adfm.202213578>
- [4] H. Wang, X.T. Xu, X.Y. Gao, Y.Q. Li, T. Lu, L.K. Pan, “Design of three-dimensional faradic electrode materials for high-performance capacitive deionization,” *Coordination Chemistry Reviews*, vol. 510, 215835, 2024. DOI: <https://doi.org/10.1016/j.ccr.2024.215835>
- [5] S. Kumar, N.M. Aldaqqa, E. Alhseinat, D. Shetty, “Electrode Materials for Desalination of Water via Capacitive Deionization,” *Angewandte Chemie International Edition*, vol. 62, e202302180, 2023. DOI: <https://doi.org/10.1002/anie.202302180>
- [6] J.Q. Guo, Y. Wang, Y.M. Cai, H. Zhang, Y.L. Li, D.X. Liu, “Ni-doping Cu-Prussian blue analogue/carbon nanotubes composite (Ni-CuPBA/CNTs) with 3D electronic channel-rich network structure for capacitive deionization,” *Desalination*, vol. 528, 115622, 2022. DOI: <https://doi.org/10.1016/j.desal.2022.115622>
- [7] Y.M. Cai, W. Zhang, J.S. Zhao, Y. Wang, “Flexible structural construction of the ternary composite Ni₂Co-Prussian blue analogue@MXene/polypyrrole for high-capacity capacitive deionization,” *Applied Surface Science*, vol. 622, 156926, 2023. DOI: <https://doi.org/10.1016/j.apsusc.2023.156926>
- [8] F.F. Jia, K.G. Sun, B.Q. Yang, X. Zhang, Q.M. Wang, S.X. Song, “Defect-rich molybdenum disulfide as electrode for enhanced capacitive deionization from water,” *Desalination*, vol. 446, pp. 21–30, 2018. DOI: <https://doi.org/10.1016/j.desal.2018.08.024>
- [9] H. Wang, Y. Liu, Y. Li, X. Xu, T. Lu, L. Pan, “Tailoring the electrode material and structure of rocking-chair capacitive deionization for high-performance desalination,” *Materials Horizons*, vol. 11, pp. 5209–5219, 2024. DOI: <https://doi.org/10.1039/D4MH00773E>
- [10] X.T. Xu, M. Eguchi, Y. Asakura, L.K. Pan, Y. Yamauchi, “Metal-organic framework derivatives for promoted capacitive deionization of oxygenated saline water,” *Energy & Environmental Science*, vol. 16, pp. 1815–1820, 2023. DOI: <https://doi.org/10.1039/D2EE03530H>
- [11] H. Wang, Y. Zhu, Y. Liu, Y. Li, X. Xu, X. Liu, H. Zhao, L. Pan, “Challenges and design strategies of metal sulfides for superior-performance capacitive deionization,” *Nano Energy*, vol. 142, 111175, 2025. DOI: <https://doi.org/10.1016/j.nanoen.2025.111175>
- [12] M. Jiang, Y. Zhang, Z. Yang, H. Li, J. Li, J. Li, T. Lu, C. Wang, G. Yang, L. Pan, “A data-driven interpretable method to predict capacities of metal ion doped TiO₂ anode materials for lithium-ion batteries using machine learning classifiers,” *Inorganic Chemistry Frontiers*, vol. 10, pp. 6646–6654, 2023. DOI: <https://doi.org/10.1039/D3QI01705B>
- [13] T. Wang, R. Pan, M.L. Martins, J. Cui, Z. Huang, B.P. Thapaliya, C.-L. Do-Thanh, M. Zhou, J. Fan, Z. Yang, M. Chi, T. Kobayashi, J. Wu, E. Mamontov, S. Dai, “Machine-learning-assisted material discovery of oxygen-rich highly porous carbon active materials for aqueous supercapacitors,” *Nature Communications*, vol. 14, 4607, 2023. DOI: <https://doi.org/10.1038/s41467-023-40282-1>
- [14] G. Xu, M. Jiang, J. Li, X. Xuan, J. Li, T. Lu, L. Pan, “Machine learning-accelerated discovery and design of electrode materials and electrolytes for lithium ion batteries,” *Energy Storage Materials*, vol. 72, 103710, 2024. DOI: <https://doi.org/10.1016/j.ensm.2024.103710>
- [15] H. Wang, Y. Li, X. Xuan, K. Wang, Y.-F. Yao, L. Pan, “Machine Learning Accelerated Discovery of Covalent Organic Frameworks for Environmental and Energy Applications,” *Environmental Science & Technology*, vol. 59, pp. 6361–6378, 2025. DOI: <https://doi.org/10.1021/acs.est.5c00390>
- [16] M. Jiang, Z. Yang, T. Lu, X. Liu, J. Li, C. Wang, G. Yang, L. Pan, “Machine learning accelerated study for predicting the lattice constant and substitution energy of metal doped titanium dioxide,” *Ceramics International*, vol. 50, pp. 1079–1086, 2024. DOI: <https://doi.org/10.1016/j.ceramint.2023.10.201>
- [17] G. Xu, Y. Zhang, M. Jiang, J. Li, H. Sun, J. Li, T. Lu, C. Wang, G. Yang, L. Pan, “A machine learning-assisted study on organic solvents in electrolytes for expanding the electrochemical stable window of zinc-ion batteries,” *Chemical Engineering Journal*, vol. 476, 146676, 2023. DOI: <https://doi.org/10.1016/j.cej.2023.146676>
- [18] H. Wang, Y. Zhu, J. Li, X. Liu, Y. Ma, Y. Yao, J. Zhang, L. Pan, “Machine Learning-Accelerated Exploration on Element Doping-triggering Material Performance Improvement for Energy Conversion and Storage Applications,” *Journal of Materials Chemistry A*, vol. 13, pp. 17197–17213, 2025. DOI: <https://doi.org/10.1039/D5TA00922G>
- [19] F. Saffarimandoab, R. Mattesini, W. Fu, E.E. Kuruoglu, X. Zhang, “‘Insights on features’ contribution to desalination dynamics and capacity of capacitive deionization through machine learning study,” *Desalination*, vol. 515, 115197, 2021. DOI: <https://doi.org/10.1016/j.desal.2021.115197>
- [20] S. Park, A.T. Angeles, M. Son, J. Shim, K. Chon, K.H. Cho, “Predicting the salt adsorption capacity of different capacitive deionization electrodes using random forest,” *Desalination*, vol. 537, 115826, 2022. DOI: <https://doi.org/10.1016/j.desal.2022.115826>
- [21] Y. Zhu, B. Lian, Y. Wang, C. Miller, C. Bales, J. Fletcher, L. Yao, T.D. Waite, “Machine learning modelling of a membrane capacitive deionization (MCDI) system for prediction of long-term system performance and optimization of process control parameters in remote brackish water desalination,” *Water Research*, vol. 227, 119349, 2022. DOI: <https://doi.org/10.1016/j.watres.2022.119349>
- [22] S.H. Chae, S.W. Hong, M. Son, “Metadata and feature importance analyses of membrane capacitive deionization models: is a water treatment artificial intelligence panacea possible?” *Desalination*, vol. 585, 117784, 2024. DOI: <https://doi.org/10.1016/j.desal.2024.117784>
- [23] H. Wang, M. Jiang, G. Xu, C. Wang, X. Xu, Y. Liu, Y. Li, T. Lu, G. Yang, L. Pan, “Machine Learning-Guided Prediction of Desalination Capacity and Rate of Porous Carbons for Capacitive Deionization,” *Small*, vol. 20, 2401214, 2024. DOI: <https://doi.org/10.1002/smll.202401214>
- [24] H. Wang, Y. Li, Y. Liu, X. Xu, T. Lu, L. Pan, “Advancement of capacitive deionization propelled by machine learning approach,” *Separation and Purification Technology*, vol. 354, 129423, 2025. DOI: <https://doi.org/10.1016/j.seppur.2024.129423>
- [25] H. Wang, C. Wang, J. Li, Y. Li, Y. Liu, Z. Chen, X. Liu, G. Yang, L. Pan, “A data-driven machine learning approach to predict desalination

- capacities of faradic materials for capacitive deionization,” *Separation and Purification Technology*, vol. 376, 134181, 2025. DOI: <https://doi.org/10.1016/j.seppur.2025.134181>
- [26] B.Z. Xia, Z.P. Qin, H.Y. Fu, “Rapid estimation of battery state of health using partial electrochemical impedance spectra and interpretable machine learning,” *Journal of Power Sources*, vol. 603, 234413, 2024. DOI: <https://doi.org/10.1016/j.jpowsour.2024.234413>
- [27] A.T. Le, A. Pandey, R. Sirohi, P. Sharma, W.H. Chen, N.D.K. Pham, V.D. Tran, X.P. Nguyen, A.T. Hoang, “Precise Prediction of Biochar Yield and Proximate Analysis by Modern Machine Learning and SHapley Additive exPlanations,” *Energy & Fuels*, vol. 37, pp. 17310–17327, 2023. DOI: <https://doi.org/10.1021/acs.energyfuels.3c02868>
- [28] S. Tan, R. Wang, G. Song, S. Qi, K. Zhang, Z. Zhao, Q. Yin, “Machine learning and Shapley Additive Explanation-based interpretable prediction of the electrocatalytic performance of N-doped carbon materials,” *Fuel*, vol. 355, 129469, 2024. DOI: <https://doi.org/10.1016/j.fuel.2023.129469>
- [29] X. Li, Z. Huang, S. Shao, Y. Cai, “Machine learning prediction of physical properties and nitrogen content of porous carbon from agricultural wastes: Effects of activation and doping process,” *Fuel*, vol. 356, 129623, 2024. DOI: <https://doi.org/10.1016/j.fuel.2023.129623>
- [30] F. Saffarimiandoab, R. Mattesini, W. Fu, E.E. Kuruoglu, X. Zhang, “Interpretable machine learning modeling of capacitive deionization for contribution analysis of electrode and process features,” *Journal of Materials Chemistry A*, vol. 9, pp. 2259–2268, 2021. DOI: <https://doi.org/10.1039/D0TA09531A>